MILITARY SPECIFICATION

POLYURETHANE FOAM KIT, RIGID

This specification is approved for use by Field Command Defense Nuclear Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a rigid polyurethane foam kit for use in certain nuclear ordnance Explosive Ordnance Disposal operations. The system shall be a three-component liquid, foam-in-place type.

2* APPLICABLE DOCUMENTS

2.1 Issues of documents. The following document of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packaging

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D93-73 - Flash Point By Pensky-Martens Closed Tester

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z129-1-1976 - Precautionary Labeling of Hazardous Industrial Chemicals

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2 and 6.3), the contractor shall furnish sample unit(s) for first article inspection per MIL-P-26514 (see 4.6) for Type II, Class 1 foams.

3.2 Materials. The materials used in production of the polyurethane foam kits shall consist of a two component liquid system (an isocyanate component and a polyol component). These components will meet the requirements of Type II, Class 1 foam per MIL-P-26514 and those requirements specified herein. The third component will be a delayed-action catalyst, composed of a tertiary amine salt in a dipropylene glycol solvent. to provide for low temperature performance of the system.

3.2.1 Toxicity . The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate departmental medical service who will act as an advisor to the procuring agency.

3.2.2 Formulation. Formulation of the components shall conform to the following ratio. The catalyst will be preblended with the polyol component.

Isocyanate component	100.0 parts
Polyol component	99.5 parts
Catalyst	0.5 parts

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3.2.2.1 Operating temperatures. The components will perform satisfactorily in the ambient temperature range of 0°F to 100°F (-17.8°C to 37.8°C) , when the components are in the temperature range of 20°F to 100°F (-6.7°C to 37.8°C).

3.2.2.2 Preconditioning temperatures. Ambient temperature range of performance will be extended to -20°F to 120°F (-28.9°C to 48.9°C) when components are preconditioned by heating or cooling to 72°F (22.2°C) for application to cold or hot hardware respectively.

3.2.3 Properties. Measured at room temperature 68°F (20°C) unless otherwise specified.

3.2.3.1 Isocyanate and polyol components. Per MIL-P-26514 and as specified below.

3.2.3.1.1 Cream time. 90 sec.

3.2.3.1.2 Rise time. 360 to 370 sec.

3.2.3.1.3 Tack free time. 300 sec.

3.2.3.1.4 Density. 2 lbs per cu. ft.

3.2.3.1.5 Compressive strength. 30 psi.

3.2.3.2 Catalyst component. The delayed action catalyst to be blended with the polyol component will have the following properties. The properties are provided for definition purposes. This component shall not be subject to separate conformance testing prior to blending with the polyol component. Properties are measured at 80°F (26.7°C) unless otherwise specified.

a. Viscosity. 66 cps

b. Density. 1.125 lbs per cu. ft.

c. pH. 3.85 (in a 2% aqueous solution)

d. Boiling point. 330°F (166°C)

e. Water volubility. 1 to 1

f. Specific gravity. 1.1086 (measured at 77°F (25°C))

g. Flash point. 155°F {68.3°C) when measured using ASTM D93-73

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h. Toxicity. When tested for acute toxicity, primary skin and eye irritation per the Federal Hazardous Substances Act (CFR 1500.3, 1500.40, 1500.41 and 1500.42) will not attain a toxic rating requiring labeling per ANSI 2129-1-1976.

3.2.4 Shelf life. The minimum shelf life of the components shall not be less than two years beginning at the time of delivery when stored in their original containers at a temperature between 50°F and 95°F (10°C and 35°C). Criteria for acceptable performance after two years storage will be no more than 5 percent (5 sec) increase in cream time, no more than 20 percent (60 sec) increase in tack free time, and no more than 25 percent (0.5 lb) increase in density (see 4.6.1).

3.2.5 Instructions. The supplier shall mark each unit container with mixing instructions per MIL-P-26514 (see 5.3.1).

3.2.6 Identification. The manufacturer shall identify each component container per MIL-P-26514 (see 5.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.6)

b. Quality conformance inspection (see 4.7)

4.3 Toxicological data and formulation. The contractor shall furnish the toxicological data and formulation required to evaluate the safety of the material for the proposed use.

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-P-26514 for Type II, Class 1 polyurethane foams. The catalyst component will be preblended with the polyol component prior to testing.

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4.4.1 Test specimens. Will be prepared in accordance with MIL-P-26514.

4.5 Sampling and container inspection. Will conform to those procedures specified in MIL-P-26514.

4.6 First article inspection. If required, will conform to those procedures specified for first article inspection of Type II, Class 1 polyurethane foams in MIL-P-26514. First article inspection may be waived when the conditions for waiver described in MIL-P-26514, have been met. When conducted, the designated first article test report will be prepared.

4.6.1 Shelf life. Manufacturers shall submit certificates that components will meet the requirements of MIL-P-26514 after storage for one year, beginning at the time of delivery, in their original containers at a temperature between 50°F and 95°F (10°C and 35°C). This certificate will also certify that those components will meet the special shelf life requirements of this specification for density, cream time and tack free time after a storage period of two years (see 3.2.4).

4.7 Quality conformance inspection. Except as otherwise specified, quality conformance inspection will be limited to those acceptance tests required by MIL-P-26514 for Type II Class 1 foams. For those tests the polyol component and the catalyst component will be preblended in the ratios designated (see 3.2.1).

4.8 Methods of inspection. The test methods for the properties of the Type II, Class 1 isocyanate component and polyol component of polyurethane foams specified for the first article inspection and acceptance tests per MIL-P-26514 will be used except as otherwise indicated. Test conditions designated will be followed.

5. PACKAGING

5.1 Preservation-packaging. Except as specified below preservation-packaging shall conform to MIL-P-26514, Type II polyurethane foams.

5.1.1 Kit packaging. Polyurethane foam kits will be prepared by placing 107 grams of the isocyanate component into a high quality 4 oz wide-mouth, linear-polyethylene bottle with a leakproof screw cap closure of 28 mm diameter (min); and 100 grams of the polyol component/catalyst component blend into a high quality 8 oz wide-mouth, linear-polyethylene bottle with a leakproof screw cap closure of 38 mm diameter (rein).

5.1.2 Bulk packaging. Packaging of both the isocyanate component and the blended polyol/catalyst components, in clean, closed head drums, per MIL-P-26514, may be specified by the procuring activity for later conversion into kit form.

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5.1.3 Air purge. To maximize shelf life, the air space over the liquids in the kits and in bulk containers will be flushed out with nitrogen before capping.

5.2 Packing. Shall conform to MIL-P-26514 for Type II liquid components.

5.3 Marking. Shall conform to MIL-P-26514 (see 3.2.6).

5.3.1 Labels and color coding. Each container will contain the water-proof labels and color coding required for Type II polyurethane foams specified in MIL-P-26514. Labels will include item part number (see 6.3).

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. Polyurethane foam kits covered by this specification are intended for use in designated explosive ordnance disposal operations to immobilize tubing and loose parts, as well as stabilize cracked or crumbled high explosive, in designated temperature ranges, for nuclear ordnance weapon systems.

6.2 Ordering data. The following ordering data shall be specified for kits conforming to this specification.

a. Title, number and date of this specification

b. Hydrolytic stability - state if not required (see 3.2.3.1)

c. Selection of applicable level of packaging (see 5.1)

d. Selection of applicable level of packing (see 5.2)

e. Marking (see 5.3)

f. Indicate whether first article inspection is waived (see 4.6)

g. Mix ratios. 1 to 1

h. Preconditioning temperatures (see 3.2.2.2)

i. Capacity (see 5.1)

j. Container color coding, if other than as specified (see 5.3.1)

6.3 Part number. The part number assigned to the product prepared in accordance with this specification will be M39500-1.

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6.4 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a sample selected from the first ten production items. The first article should consist of two kits or two quarts of each component of bulk material. Other instructions for first article inspection will conform to MIL-P-26514 (see 3.1 and 4.6).

Custodians:

DOD - DS

Preparing Activity:

DOD - DS

(Project 1145-0002)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-39500 (DS) Basic

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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MIL-P-3950D(DS)
AMENDMENT 2
1 June 1982
SUPERSEDING
AMENDMENT 1
16 November 1979

MILITARY SPECIFICATION

POLYURETHANE FOAM KIT, RIGID

This amendment forms a part of Military Specification MIL-P-39500(DS), dated 5 September 1979, and is approved for use by Field Command Defense Nuclear Agency and is available for use by all Departments and Agencies of the Department of Defense.

2.1 Following "MIL-P-26514," insert:

DRAWINGS

ARMY ARMAMENT R & D COMMAND

11831301 - Urethane Foam

PAGE 3

3.2.3.1 Add new paragraph. . . . . .

3.2.3.1.6 Initiation time. 45 sec (after a 20 sec. mix).

PAGE 4

3.2.4 line 2: Delete "one year" and substitute "two years."

* line 4: Following first sentence, delete entire second sentence.

PAGE 5

* 4.6.1 lines 4 to 7: Following first sentence, insert: "This certificate will also certify that those components will meet the special shelf life requirements of this specification for density, cream time and tack free time after a storage period of two years (see 3.2.4)."

PAGE 6

6.3 line 2: Following "M9500-1," insert: "(Army part number 11831301)".

NUOR

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NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, correctionns, deletions) from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
DOD-DS

Preparing activity:
DOD-DS

(Project NUOR-0015)

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NOTICE OF ACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-P-39500 (AR)
NOTICE 1
21 June 1995

MILITARY SPECIFICATION

POLYURETHANE FOAM KIT, RIGID

This notice should be filed in front of MIL-P–39500 (AR), dated 05 September 1979

MIL-P-39500 (AR) dated 05 September 1979, with Amendment 2 dated, 01 June 1982 is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity
Army-AR

AMSC N/A

FSC 1195

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